

ECON 0100 | Vignette B3 | Equilibrium

I Solutions

Answers and work are shown in the red blocks.

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve:

$$P_d = 14 - \frac{1}{2}Q_d$$

Pumpkin pasties are produced by many sellers according to the following supply curve:

$$P = 2 + \frac{1}{2}Q_s$$

Prices are in galleons and quantity is in pasties.

Q1 | Equilibrium

Find and plot the equilibrium price and quantity for pumpkin pasties.

I $Q^* = 12$ pasties at $P^* = 8$ galleons.

Plot. Demand: vertical intercept 14, horizontal intercept $0 = 14 - \frac{1}{2}Q_d \rightarrow Q_d = 28$. Supply: vertical intercept 2, rising by 1 galleon for every 2 pasties.

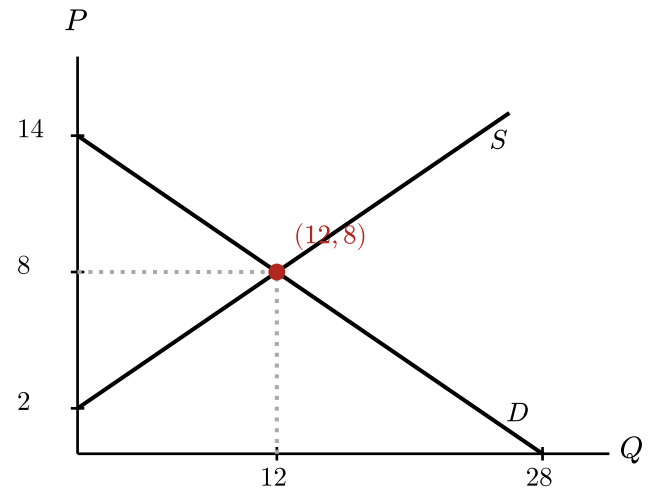
Equilibrium. At equilibrium $Q_s = Q_d = Q^*$, so set the two curves equal:

$$2 + \frac{1}{2}Q^* = 14 - \frac{1}{2}Q^* \rightarrow Q^* = 12$$

Plug Q^* back into either curve:

$$P^* = 2 + \frac{1}{2} \cdot 12 = 2 + 6 = 8$$

Star the pair (12, 8) where the curves cross.



Q2 | Surplus

Find and plot the consumer and producer surplus in the market equilibrium you found in Q1.

CS = 36 galleons. PS = 36 galleons.

Consumer surplus is the triangle below demand and above P^* , out to Q^* :

$$CS = h \cdot b \cdot \frac{1}{2} = (14 - 8) \cdot 12 \cdot \frac{1}{2} = 36$$

Producer surplus is the triangle above supply and below P^* , out to Q^* :

$$PS = h \cdot b \cdot \frac{1}{2} = (8 - 2) \cdot 12 \cdot \frac{1}{2} = 36$$

Total surplus is $36 + 36 = 72$ galleons.

